Apoorv Mishra

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Experience

CodeAlpha **Jun 2026 – Jul 2026**
Artificial Intelligence Intern **Remote**

- Built a **real-time object detection and tracking dashboard** in **Python, OpenCV, YOLOv8, and ByteTrack**, showing live object counts and cumulative unique-object totals over the video feed.
- Added **automatic .mp4 recording** of the tracked feed and developed a **language translation tool** as part of the internship project set.
- Published all tasks in a public GitHub repository and earned an **internship completion certificate**.

UPES ACM Student Chapter **Aug 2026 – Present**
Technical Core Member **Dehradun, India**

- Contribute to **software projects**, including the **React and Vite** website for the chapter (upesacm.org); help organize **technical events** and **mentor peers** in **C and Data Structures**.

Projects

Quanta CareTrace | *Agentic clinical documentation, Tech Zephyr 4.0 (team of 4, backend)* GitHub

- Built backend services for a workflow that drafts follow-up notes and **blocks unsafe prescription candidates** with a deterministic guideline rule (e.g., Lisinopril with potassium above 5.0 mEq/L), keeping the LLM advisory only.
- Imported **20,000 synthetic patient records** into SQLite with paginated search and **50 patient-linked case workspaces**.
- Added **server-side role checks** (only clinicians approve), a **per-step audit log**, and a **14-check acceptance suite**.
- **Stack:** Python, SQLite, Gemini, Ollama, Server-Sent Events

Hotel & Amenity Management System | *Streamlit app, team of 2* GitHub

- Built a **Streamlit** app on **MongoDB** with object-oriented CRUD modules for hotels, rooms, customers, bookings, and amenities.
- Used **Python threading and locks** to prevent **double-booking** when multiple users reserve the same room.
- **Stack:** Python, Streamlit, MongoDB, PyMongo

Multithreaded Terminal Air Hockey | *Real-time two-player game in C* GitHub

- Designed a **three-thread architecture** (input, game logic, rendering) sharing state behind a **mutex**, with physics ticking about **15 times per second**.
- Slept the input thread for **10 ms** to stop busy-spinning and keep idle CPU usage low.
- **Stack:** C, POSIX Threads, termios, ANSI escape codes

Reborn | *INNOV8 4.0, IIT Delhi (Top 40 finalist)* GitHub

- Compared three ensemble ranking approaches using Hits@150 (best **100/150**) and produced **500 ranked candidates in about 110 seconds**.
- Wrote **decision-logic modules** (normalization, deduplication, fraud checks, scoring, LLM notes) for an autonomous agent in the finals' live simulation.
- **Stack:** Python

The Quiet Keep | *Side-scrolling platformer, plain JavaScript* Live demo

- Built a **ten-world platformer from scratch** on HTML5 Canvas with sword combos, fireballs, a village shop, quests and a guardian boss in each world.
- Deployed on **Vercel** with desktop and touch controls, so anyone can play it in the browser.
- **Stack:** JavaScript, HTML5 Canvas

Education

University of Petroleum and Energy Studies **Expected May 2029**
B.Tech in Computer Science and Engineering (AI/ML) | CGPA: 8.1/10.0 **Dehradun, India**
Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Systems Programming

Technical Skills

Languages: Python, C, JavaScript
Frameworks & Libraries: Streamlit, React, OpenCV, YOLOv8 (Ultralytics), NumPy
Databases: MongoDB (PyMongo, Atlas), SQLite
Tools & Concepts: Git, GitHub, Linux Shell, Multithreading & Synchronization, OOP, Data Structures & Algorithms

Achievements

Top 40 Finalist | IIT Delhi INNOV8 4.0
Solved 197 problems on LeetCode | Dynamic Programming, Trees, Graphs